

CS & IT ENGINEERING



Operating System

Process Synchronization

Lecture -1

By- Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

doubt?
Multithreading

Topic

System Call: Fork()

Topics to be Covered



Topic

Synchronization

Topic

Race Condition

#Q. Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

GATE - 2006

0%



✓ 10.6%



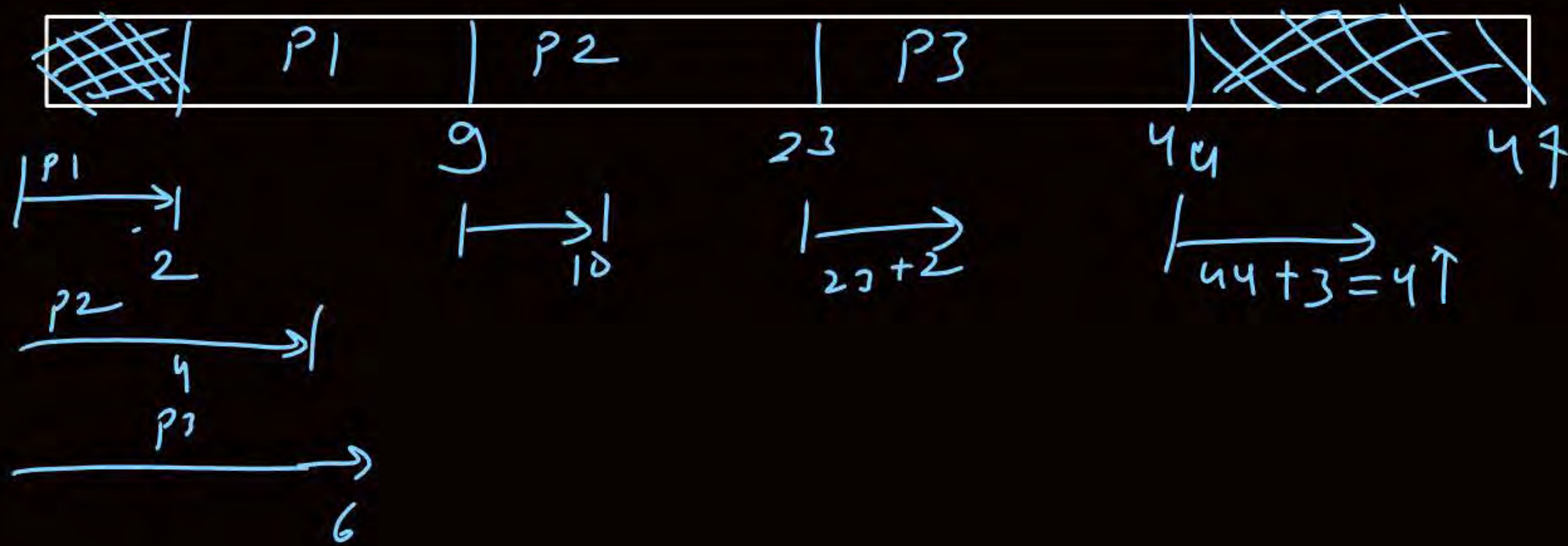
30.0%



89.4%

		I/O	CPU	I/O
P ₁	10	2 _x	7 _x	1 _x
P ₂	20	4 _x	14 _x	2 _x
P ₃	30	6 _x	21 _x	3 _x

$$\frac{5}{47} * 100\%$$



#Q. Consider four processes P, Q, R, and S scheduled on a CPU as per round-robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

GATE-2022

(A) $P = 4, Q = 10, R = 6, S = 2$

(B) $P = 2, Q = 9, R = 5, S = 1$

(C) $P = 4, Q = 12, R = 5, S = 4$

☒ (D) $P = 3, Q = 7, R = 7, S = 3$



$P \Rightarrow 3T$
 $P \Rightarrow 4$

$Q \Rightarrow 9 \text{ to } 12$

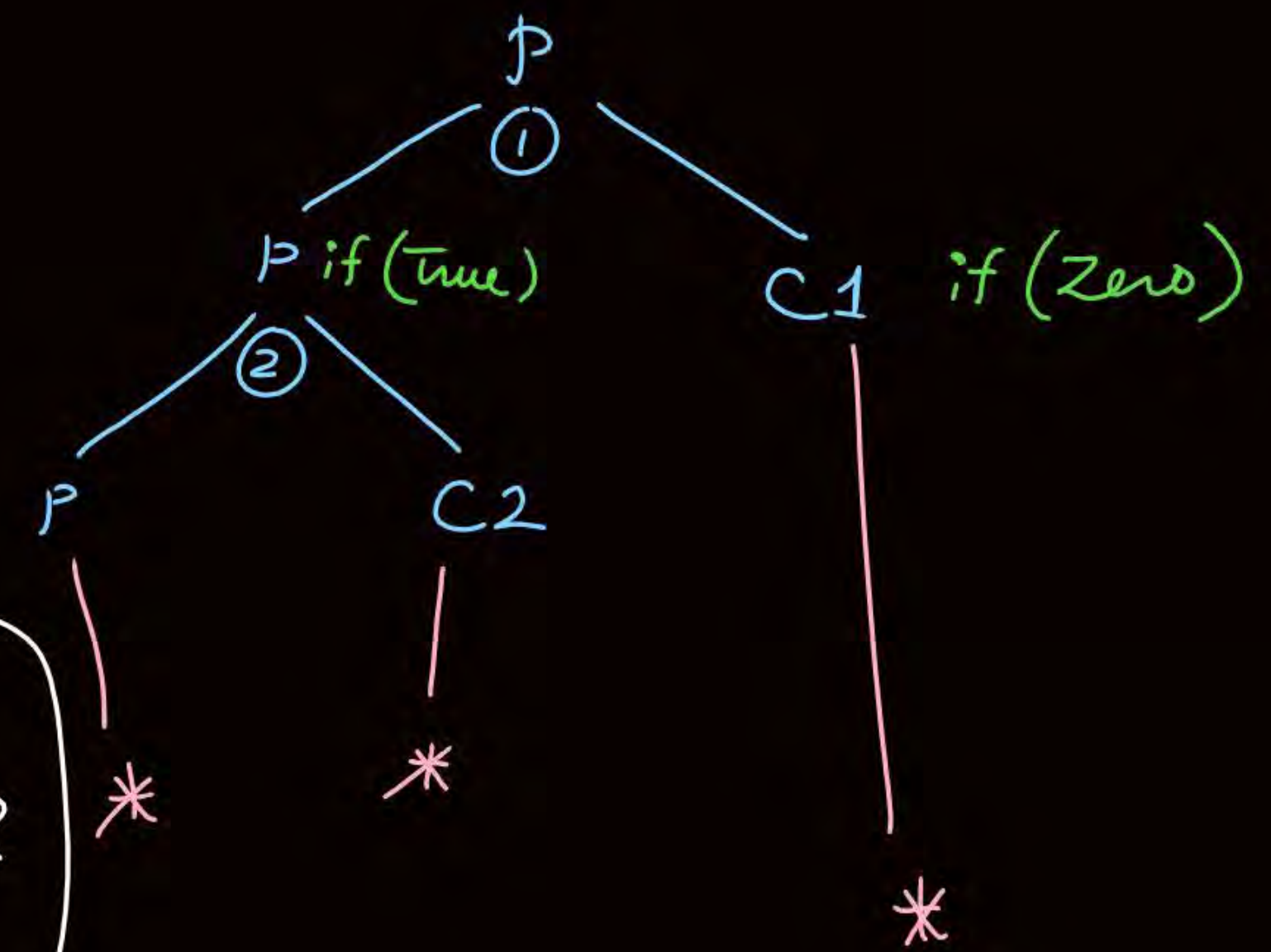
$R \Rightarrow 5 \text{ to } 8$

$S \Rightarrow 1 \text{ to } 4$

Ques)

```
if(fork①())  
{  
    fork②();  
}  
printf("*");
```

no. of times * printed 3?



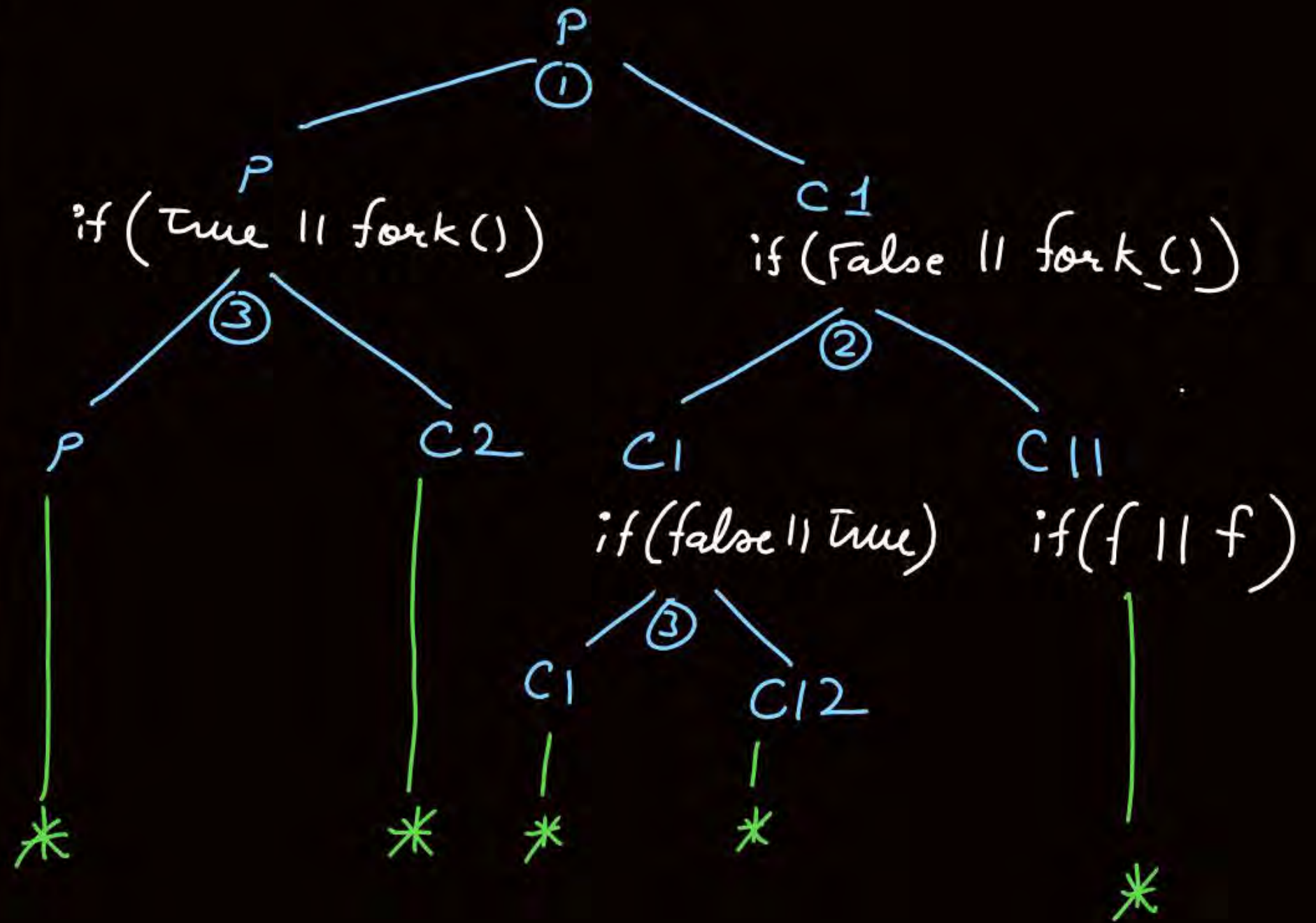
Ques)

¹
if (fork() || fork(²))

{
 ³
 fork();

}
printf("*");

no. of times * printed 5?



H.W.

Ques)

```
if (fork() && fork())
```

```
{  
    fork();
```

```
}
```

```
printf("*");
```

no. of times * printed _____ ?



Topic : Wait() System Call



A call to wait() blocks the calling process until one of its child processes completes

After child process terminates, parent continues its execution after wait system call instruction



#Q. Consider the following code snippet using the `fork()` and `wait()` system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors.

```
int x = 3;  
while(x > 0) {  
    fork();  
    printf("hello");  
    wait(NULL);  
    x--;  
}
```

The total number of times the `printf` statement is executed is _____?

GATE-2024

New chapter

Process synchronization



Topic : Process Types

1. Independent $\rightarrow$ no communication with any other process.
2. Cooperating/Coordinating/Communicating $\rightarrow$ do communicate with other processes.

OS does the IPC work

$\downarrow$
interprocess communication



Topic : Need Of Synchronization

↓
b/w communicating process





Topic : Problems Without Synchronization

Problems without Synchronization:

- Inconsistency
- Loss of Data
- Deadlock





Topic : Race Condition

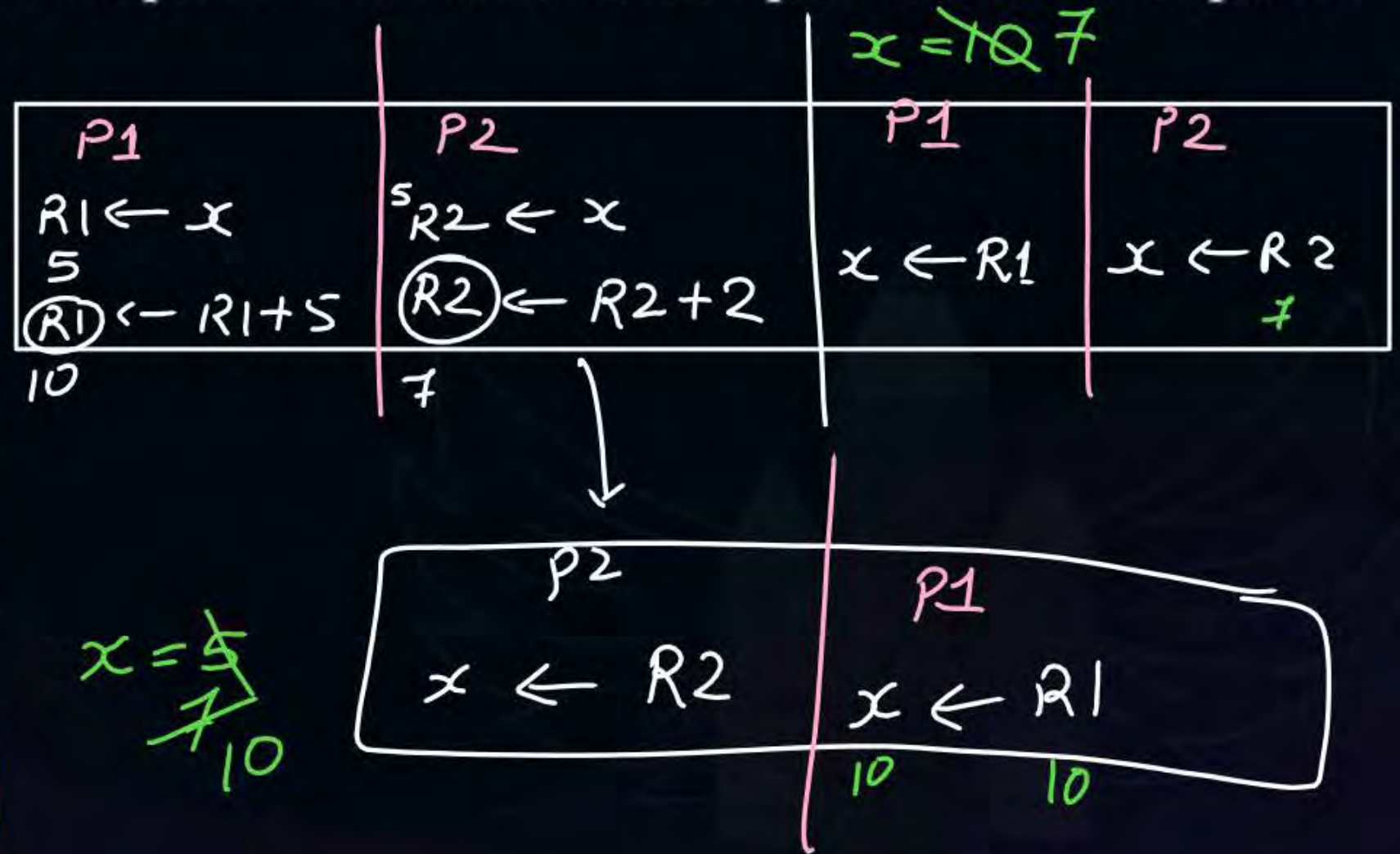


A race condition is an undesirable situation, it occurs when the final result of concurrent processes depends on the sequence in which the processes complete their execution.

<u>P1</u>	<u>P2</u>
$R1 \leftarrow x$	$R2 \leftarrow x$
$R1 \leftarrow R1 + 5$	$R2 \leftarrow R2 + 2$
$x \leftarrow R1$	$x \leftarrow R2$

$x = 5$

expected
 $x = 12$



$$x = 10$$

P1

$$R1 = x^{10}$$

$$R1 = R1 * 2$$

$$x = R1$$

P2

$$R2 = x^{10}$$

$$R2 = R2 + 3$$

$$x = R2$$

Distinct possible values of x $\frac{4}{1}$?

13, 20, 23, 26

Case 1:-

P1 process completed then P2 starts.

$$x = \cancel{10} \cancel{20} 23$$

Case 2:-

P2 process completed then P1 starts.

$$x = \cancel{10} \cancel{13} 26$$

Case 3:-

P1 and P2 concurrently read $x = 10$ then P1 finishes last

$$x = \cancel{10} \cancel{13} 20$$

Case 4:-

P1 and P2 concurrently read $x = 10$ then P2 finishes last.

$$x = \cancel{10} \cancel{20} 13$$

[NAT]

#Q. $X = 10$

P1

$X = X / 2$

P2

$X = X + 4$

Case 1:-

$$X = \cancel{8} 9$$

Case 2:-

$$X = \cancel{10} \cancel{14} 7$$

Case 3:- $X = 5$

Case 4:- $X = 14$

How many different values of X are possible after both processes finish executing?

Ans = 4

#Q. The following pair of processes share a common variable X.

Process A	Process B
int Y; Y = X*2; X = Y;	int Z; Z = X+1; X = Z;

X is set to 4 before either process begins execution. As usual, statements within a process are executed sequentially, but statements in process A may execute in any order with respect to statements in process B. How many different values of X are possible after both processes finish executing?



2 mins Summary

Topic

Synchronization

Topic

Race Condition



Happy Learning

THANK - YOU

